

The technical potential for onshore wind in Wallonia

A land-eligibility analysis against the Walloon regulatory framework,
and what it implies for the capacity limit used in PyPSA-Wal

Sylvain Quoilin

Integrated and Sustainable Energy Systems, University of Liège

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Abstract

PyPSA-Wal currently caps Walloon onshore wind at 6 500 MW. That number is an expert judgement taken from the PNEC wallon and from EDORA; it is not derived from a land-eligibility calculation, and the model’s own calculation — the one PyPSA-Eur performs with **atlite** — was never run against Walloon law. This report closes that gap. It builds a stand-alone workflow that computes the Walloon onshore-wind potential from the Walloon Region’s own regulatory cartography: the plan de secteur, the *cadre de référence éolien* of 24 January 2024, the Natura 2000 network, the nature reserves, the landscape inventories and the natural-hazard maps, all retrieved live from the Géoportail de la Wallonie. The constraint set is applied in a documented ladder so that the cost of each rule in land and in megawatts is visible separately.

Against the reference constraint set (S5_risk) and the reference turbine class (NREL 2020ATB 4.0 MW (tip 185 m)), 870 km² of the Walloon Region survive every legal exclusion — 5.14 % of the territory — supporting 4 417 MW at 5.08 MW/km². Restricted to the region the model actually calls BEWAL, the figure is 4 127 MW. Across turbine classes and array-spacing assumptions the BEWAL result spans 2 327–7 059 MW.

Four results matter more than the headline number.

Two independent methods converge on 2–4.5 GW. Only 5.14 % of Wallonia survives the legal constraints, and that land is fragmented into 5 290 patches of median size 3 ha. Screening it down to patches that could actually hold a turbine array gives 2 066 MW — against the 2 559 MW implied by the 457 potential turbines of the Region’s own 2022 site-by-site simulation, a different method on different data. The unscreened figure is 4 417 MW. The current 6 500 MW cap sits above both, and is reachable in this land envelope only at a capacity density near 6.8 MW km^{−2} with no developability screen at all.

The binding constraint is settlement, not nature. Adding the setbacks of the *cadre de référence* to the plan-de-secteur zoning removes 84 % of the admissible land in one step. Everything that follows — Natura 2000, reserves, landscape perimeters, flood, karst, water capture — together removes 529 km² more.

PyPSA-Eur’s default land analysis is not usable here. It returns 18 270 MW for the same region, because it admits broad-leaved forest, ignores the plan de secteur entirely, and cannot see scattered housing.

The model’s BEWAL node is not Wallonia. 2 191 km² of Walloon territory — western Hainaut, 13 % of the Region — is assigned to the Flemish node. It carries 6.6 % of the eligible land, so the resource effect is modest; the correctness problem is not, and it applies to every land-based Walloon quantity in the model.

No input of PyPSA-Wal has been modified. This document is the evidence base for that decision, not the decision.

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1 Why this analysis exists

1.1 The number under review

The Walloon onshore-wind capacity limit in PyPSA-Wal is a single row of `config/input_parameters_for_models.csv`:

```
potential:BEWAL:onwind:p_nom_max = 6500 MW, source “PNEC Wallon et EDORA”,
described as “potentiel technique, calculé comme la somme de ce qui est plausible d’atteindre
en 2030 comme capacité et des projets à l’étude”, with a note that EDORA estimates an
additional 3.8 GW of wind potential by 2050 on the basis of projects under environmental-
impact assessment.
```

Two things follow from that wording. It is a *deployment* expectation assembled from a pipeline of known projects, not a *land* potential; and it carries no spatial calculation that anyone can re-run or contest. For an internal scenario exercise that is acceptable. For a number that is reported to the administration as the ceiling on what Wallonia can host, it is not: the first question the cabinet will ask is “*where does 6.5 GW fit, and under which rules?*”, and the current answer is a literature reference.

1.2 What PyPSA-Eur would have said

PyPSA-Eur ships exactly the tool needed — `atlite`’s land-eligibility analysis, driven by `scripts/determine_availability_matrix.py`. It is run for every other node in the model. For BEWAL its output is overridden by the CSV row above, so its answer has never been examined. It is 18 270 MW (18.3 GW) for the model’s BEWAL region and 19 079 MW for the administrative Walloon Region.

That the override exists is defensible — Section 4 shows the default constraint set is unusable in Wallonia. That the override was never compared against a corrected calculation is the gap this report fills.

1.3 Scope, and what this report does not claim

This is a *land-eligibility* study. It answers: how much Walloon land can legally and physically host a wind turbine, and how much capacity does that land support at conventional array spacing? It does not answer: how much of that capacity will be permitted, financed and built, or in what year. A technical potential is an upper bound on the model’s search space, and that is precisely the role `p_nom_max` plays in PyPSA-Wal — the build-rate limit and the 2030 corridor, documented in `docs/renewable-potentials.md`, carry the deployment question.

The analysis deliberately stops short of site-by-site turbine placement. The 2022 SPW/Gembloux update of the Walloon favourable-zone map does place individual machines and applies landscape inter-distance and encirclement criteria on top; Section 6.4 compares the two and explains why the difference is expected and in which direction.

2 The workflow

2.1 Design

The analysis lives in `workflow_onwind_wal/`, a self-contained Snakemake workflow. It shares nothing with the main PyPSA-Wal workflow except two read-only inputs: the region shapes (`resources/regions_onshore_base_s_adm.geojson` and `resources/nuts3_shapes.geojson`) and the `atlite` ERA5 cutout. It writes only inside its own directory and into `docs/onwind_potential_wallonia/`. Running it, breaking it or deleting its outputs cannot affect a model run.

```
cd workflow_onwind_wal
snakemake -c4 download # ~20 min, one-off, Walloon open data
snakemake -c4 all      # eligibility, potentials, figures, tables
snakemake -c4 report   # this PDF
```

The stages are:

1. **Retrieve.** Nineteen reference datasets are pulled from the ESRI-REST services of <https://geoservices.wallonie.be> (CC-BY 4.0), paginated, and written as GeoPackages with a provenance sidecar recording the service, the layer, the feature count and the retrieval date (Annex B). Feature counts are checked against the counts observed on 22 September 2026 and a deviation above 20 % raises a warning: if the Region revises a dataset, the workflow says so rather than silently changing the answer.
2. **Regions.** Two polygons are built — the administrative Walloon Region (union of the 21 BE3 NUTS-3 areas) and the model’s BEWAL node — together with their difference (Section 5).
3. **Exclusion layers.** The regulatory rules are translated into geometry, once per turbine class, because most setbacks scale with tip height or rotor diameter (Section 3).
4. **Eligibility.** `atlite’s ExclusionContainer` burns every layer onto a 100 m raster in EPSG:3035 and `cutout.availabilitymatrix` aggregates the surviving share per ERA5 cell — the identical code path PyPSA-Eur uses, so the two results are comparable by construction.
5. **Potential.** Eligible area $\times$ capacity density gives installable capacity; the capacity factor is computed with the same area-and-resource-weighted layout PyPSA-Eur applies, giving full-load hours.
6. **Report.** Tables, figures and every number in this text are generated. Nothing here is typed by hand.

2.2 Validation: the workflow reproduces the model’s own profile

Before any of the Walloon constraints are applied, the workflow has to be shown to reproduce what PyPSA-Eur itself produces. Scenario `S0_pypsa_eur` run with PyPSA-Eur’s own turbine (Vestas V112, hub 80 m) on the model’s own BEWAL region is exactly that test: same exclusion logic, same CORINE raster, same Natura 2000 raster, same ERA5 cutout.

Table 1: End-to-end validation against the profile the model actually carries.

	this workflow	PyPSA-Wal’s own profile
full-load hours, 2013 weather year	2 442 h	2 425 h
deviation		0.7 %

The reference figure is the capacity factor of `resources/walloon_2013/.../profile_adm_onwind.nc`, reported in `docs/renewable-potentials.md` §10.1. Agreement here means the cutout is read correctly, the layout weighting matches PyPSA-Eur’s, and the aggregation to the node is right — so any difference in the results that follow comes from the constraint set and not from the plumbing.

2.3 Why the constraints are declared in the config, not in the code

Every exclusion is a block of `workflow_onwind_wal/config.yaml` naming the dataset, the legal instrument and the buffer. Reviewing the constraint set — which is what the administration will actually want to do — requires reading one YAML file, not the Python. Changing a rule is a config edit, and Snakemake re-derives only what depends on it.

3 The Walloon regulatory framework

3.1 Sources of law and of practice

Four instruments define where a wind turbine may stand in Wallonia:

CoDT the *Code du Développement Territorial*, which fixes the zoning categories of the *plan de secteur* and, at art. R.II.21-1, the definition of the *principales infrastructures de communication* (PIC): motorways and regional link roads, railways, and navigable waterways.

Cadre de référence éolien the reference framework for wind development. The 2013 version imposed a setback from plan-de-secteur habitat zones of four times the *mast* height — in practice about 600 m for the machines of the time — and the 2022 update of the favourable-zone map applied it as four times the *total* height. It was replaced by the framework adopted on 24 January 2024, in force since 25 April 2024, which sets that distance at **500 m plus half the total height** and applies it to *zone d'habitat*, *zone d'habitat à caractère rural*, *zone d'aménagement communal concerté* destined for housing, and *zone d'habitat vert*. The same text lowers the definition of a wind farm from five turbines to four.

Sectoral conditions and permitting practice noise limits, aviation and defence radar clearances, and the protection of the Humain radio-astronomy station and the Wideumont weather radar.

The Walloon favourable-zone cartography the 2013 map of *zones favorables* and its 2022 update by SPW and Gembloux Agro-Bio Tech, which is the Region's own operational translation of all of the above into GIS layers. Its constraint tables are the template followed here.

3.2 The integral exclusions, and how each is implemented

Table 2 lists the integral (absolute) exclusions of the Walloon favourable-zone methodology, what this workflow does with each, and where the implementation differs.

3.3 The one rule that dominates everything: the agricultural corridor

The Walloon methodology does not treat agricultural land as uniformly available. It admits it only within 1.5 km of a PIC or of an economic-activity zone, and coniferous forest only within 750 m of a PIC. This is a *concentration* rule — keep wind development along the existing structuring infrastructure rather than scattered across the open landscape — and it has no analogue anywhere in PyPSA-Eur. It is also the single largest difference between the two constraint sets, and it works in the opposite direction to PyPSA-Eur's own treatment, which *excludes* a 1 km band around transport infrastructure.

3.4 Setbacks for the three turbine classes

The turbine classes mirror the three *scénarios de base* of the 2022 favourable-zone update (total tip height 150, 180 and 210 m) so the two studies can be read side by side. The reference class is NREL 2020ATB 4.0 MW (tip 185 m): rotor 150 m, tip height 185 m, giving a habitat-zone setback of 592 m under the 2024 framework.

Caveat. Read as *4H* on the total height — the form the 2022 favourable-zone update used — the 2024 framework is the *looser* of the two for every machine considered here: 592 m against 740 m for the reference class. Results computed on the 2013 convention would therefore be lower. The 2024 rule is used because it is the one in force; the 2013 rule is available in the config (`setbacks.habitat_zone.default: cdr2013`) as a sensitivity.

Table 2: Walloon integral exclusions and their implementation. Red marks a constraint that could not be implemented from open data.

category	Walloon rule	implementation here
technical incompatibility	water bodies, extraction zones, steep slopes	plan-de-secteur <i>Eau</i> and <i>Extraction</i> classes excluded; <i>versants</i> $> 30^\circ$ from the LiDAR relief service. The > 7 % slope criterion of the 2013 map is not applied (§8)
plan de secteur	broad-leaved forest excluded; coniferous forest admissible only within 750 m of a PIC; agricultural land only within 1.5 km of a PIC or of an economic-activity zone; <i>espaces verts, naturelle, parc</i> excluded	implemented literally as a positive list on the AFFECT classes of the <i>Zones d'affectation</i> layer; the broad-leaved/coniferous split comes from CORINE, which the plan de secteur does not carry
infrastructure safety	rail 190 m; high-speed rail 140 m; roads $1.5 \times$ blade length; motorways 50 m; HV lines $1.5 \times$ rotor diameter	buffers applied to the plan-de-secteur network layers; for the reference turbine, 112 m on roads and 225 m on HV lines
hazard zones	flood-prone areas, landslide risk, karst, close water-capture protection perimeters	EU Floods-Directive overflow extent (extreme return period), karst constraints, <i>contraintes physiques liées aux éboulements</i> , and <i>zones de prévention rapprochée IIa</i>
civil aviation	airport control zones and radar/beacon interference zones	not implemented — skeyes does not publish the exclusion geometry as open data (§8)
national defence	military airspace, defence radar distances, military airfield zones	not implemented — same reason
built heritage	classified sites	not implemented
natural heritage	nature reserves, forest reserves, wetlands of biological interest, underground cavities of scientific interest, Natura 2000	all five implemented from the FAUNE_FLORE services
biodiversity	high-priority ornithological zones	not implemented — the DEMNA priority-bird layer is not public
landscape	landscape preservation perimeters	plan-de-secteur <i>périmètres d'intérêt paysager</i> and the ADESA inventory
living environment	habitat zones of the plan de secteur, setback per the cadre de référence; dwellings outside habitat zones, 400 m	cdr2024 rule, 592 m for the reference turbine; and 400 m around every one of the 1.5 million ICAR address points
scientific	Humain radio-astronomy station, Wideumont weather radar	not implemented — perimeters not published

4 PyPSA-Eur’s default constraint set is not applicable in Wallonia

PyPSA-Eur’s onshore-wind land analysis (`config/config.default.yaml`, section `renewable.onwind`) admits CORINE grid codes 12–29, 31 and 32, excludes a 1 km buffer around codes 1–6, excludes Natura 2000, and assumes 3 MW km⁻². Five features of that specification make it unsuitable here.

1. **It admits all forest.** Codes 23, 24 and 25 are broad-leaved, coniferous and mixed forest. Wallonia is roughly one third forest, and the forest is concentrated in the Ardennes — the windiest part of the Region. Walloon law excludes broad-leaved stands outright and admits coniferous stands only near a PIC. This single difference moves the answer more than any other.
2. **It cannot see scattered housing.** CORINE’s minimum mapping unit is 25 ha. Walloon *habitat dispersé* — isolated farms and ribbon development — is invisible to it, and it is exactly what the 400 m rule protects. The 1 km buffer around CORINE urban codes is not a substitute: it is generous where settlements are large and blind where they are not.
3. **It has no zoning.** The plan de secteur is the legal instrument that decides admissibility in Wallonia, and CORINE is a land-*cover* product that knows nothing about it.
4. **Its buffer around transport infrastructure has the wrong sign.** PyPSA-Eur excludes 1 km around roads and railways; the Walloon rule requires agricultural sites to be *within* 1.5 km of them.
5. **The CORINE raster shipped in the data bundle is 250 m, 2006 vintage** (`g250_clc06_v18_5.tif`). At 250 m a single pixel is 6.25 ha; at the scale of a 400 m setback that is a coarse instrument, and the land cover is twenty years old.

Finding. PyPSA-Eur’s default land analysis returns 18 270 MW for BEWAL and 19 079 MW for the administrative Region. Those figures should not be quoted as a Walloon technical potential, and the decision to override them in `input_parameters_for_models.csv` was correct. What was missing was a corrected calculation to override them *with*.

5 The model’s BEWAL node is not Wallonia

PyPSA-Eur builds onshore regions by Voronoi-partitioning each country around its substations and clipping to the country shape. The partition is electrical, not administrative, and the Belgian three-node split (BEWAL, BEVLG, BEBRU) inherits that geometry.

Table 3: The administrative Walloon Region against the model’s BEWAL node.

Walloon Region (union of BE3 NUTS-3 areas)	16 905 km ²
PyPSA-Wal BEWAL onshore region	15 150 km ²
Walloon territory assigned to another node	2 191 km ²
coverage	89.6 %

A smaller effect runs the other way: 435 km² of the BEWAL polygon lies outside Wallonia, mostly over the French border. Walloon zoning data does not exist there, so this workflow treats it as ineligible — the conservative choice, and one that slightly understates the BEWAL figure.

The missing territory is not scattered: it is the whole of western Hainaut — Tournai, Mouscron, Ath and the Tournaisis — which sits on the Flemish side of the partition (Figure 1). That area is flat, agricultural, well served by motorways and railways, and among the windiest land in Wallonia — the kind of territory the Walloon agricultural-corridor rule favours.

It is worth comparing the two shares directly, because they do not match and the mismatch is informative: the excluded territory is 13 % of Wallonia’s surface but carries only 6.6 % of its eligible land. Western Hainaut is therefore *less* endowed per square kilometre than Wallonia on average, and the reason is the same one that dominates the whole study — the Tournaisis is

densely and continuously settled, so the 400 m dwelling setback eats most of it. The resource effect of the region mismatch is consequently modest. The correctness problem is not: a number computed on one region and applied to another is wrong regardless of how much it moves.

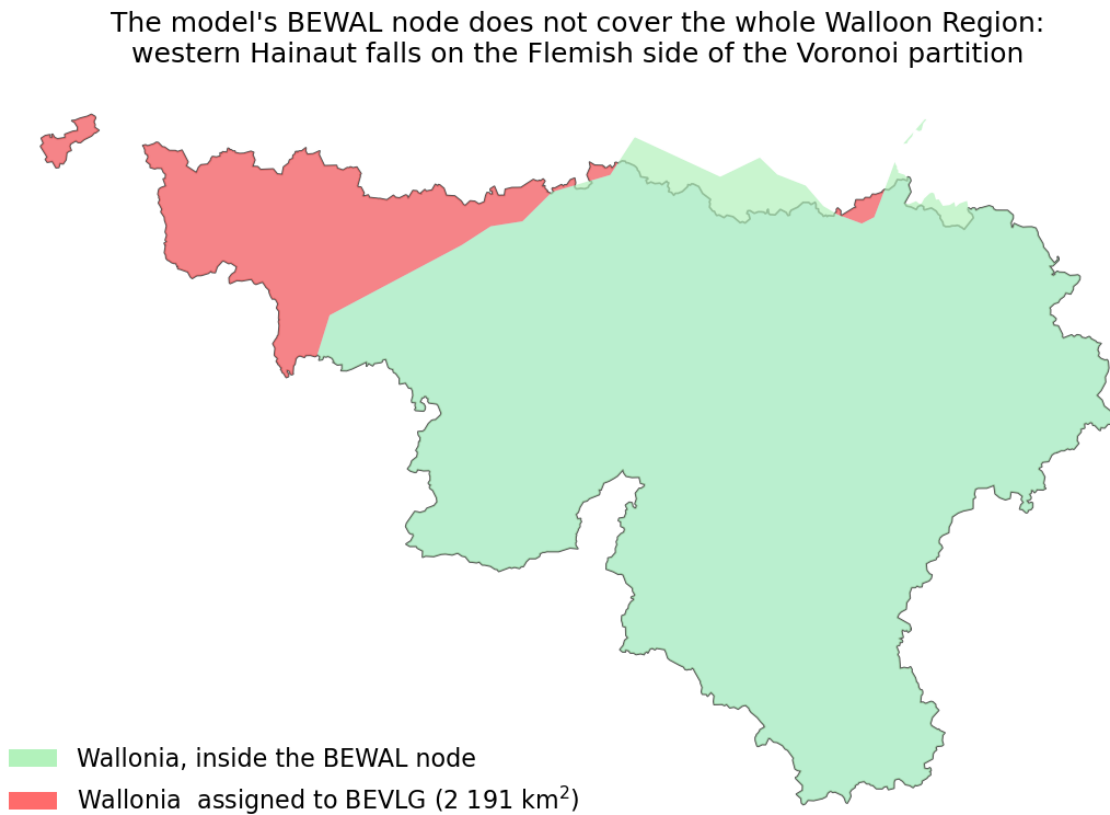


Figure 1: Wallonia and the model's BEWAL node. The red area is Walloon territory attributed to BEVLG by the Voronoi partition.

Finding. Any Walloon potential reported to the administration must state which of the two regions it refers to. A cap derived from the administrative Region and applied to the BEWAL node silently grants Wallonia the resource of land the model has already given to Flanders — and conversely, a BEWAL figure understates what Wallonia as a Region can host. Both are reported throughout this document.

Caveat. This is a defect of the model's regionalisation, not of the potential calculation, and it affects every Walloon quantity in PyPSA-Wal that is land-based — solar ground-mounted potential and biomass among them. Fixing it means correcting the busmap, which is out of scope here but should be raised separately.

6 Results

6.1 How much land each rule removes

Scenarios S1–S5 are cumulative: each adds layers to the previous one, so the difference between two consecutive rows is the cost of the rules that were added. S0 is *not* part of that sequence — it is PyPSA-Eur's own specification, a different constraint set rather than a subset of S1 — and it is shown apart in the table and in grey in the figure. It is larger than S5 but smaller than

S1, because it applies a 1 km urban buffer and Natura 2000 that S1 does not, while ignoring the zoning, setback, landscape and hazard rules that S1 and its successors impose.

Finding. The Walloon plan de secteur on its own leaves 8 521 km² admissible. Adding the setbacks of the cadre de référence — 592 m from habitat zones and 400 m from every dwelling — removes 7 122 km², or 84 % of it, in a single step. Everything that follows (protected areas, landscape perimeters, natural hazards) together removes about a third of what is left. The Walloon onshore-wind potential is governed by settlement pattern, not by nature protection and not by the wind.

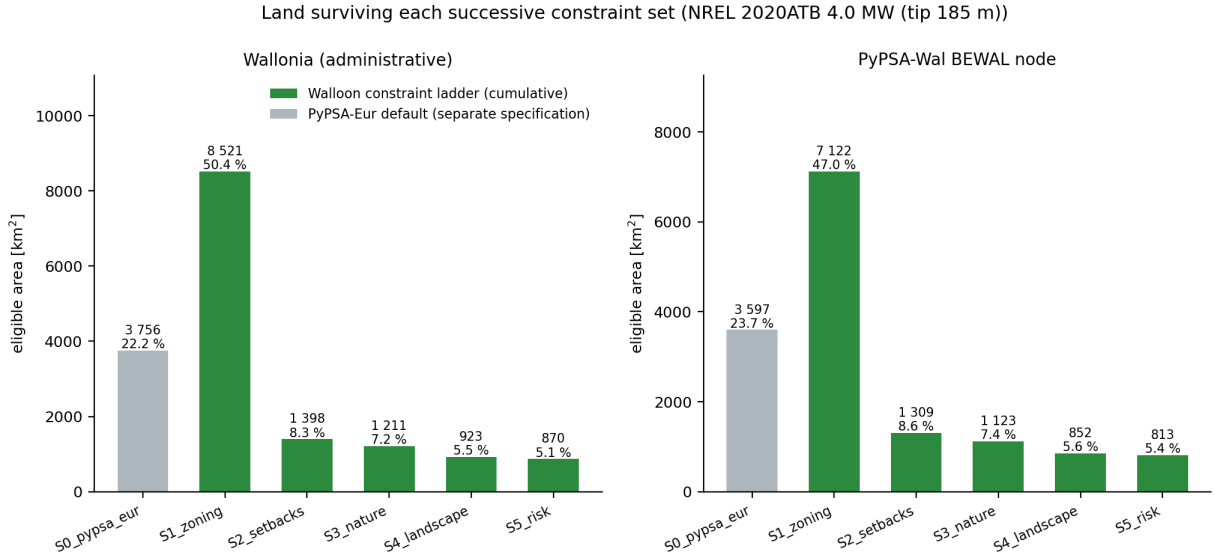


Figure 2: Land surviving each successive constraint set, for the reference turbine class.

6.2 Installable capacity

Turbine class barely moves the energy, only the capacity. Across the three classes the energy yield per square kilometre of eligible land is 16.5–17.1 GWh/km², a spread of a few per cent, while the installed capacity on the same land spans 4 257 MW to 6 457 MW. That is the expected physics: at fixed rotor-diameter spacing the swept area per square kilometre is a constant, so a low-specific-power machine trades megawatts for full-load hours at roughly constant energy. It also means the choice of turbine class is close to irrelevant for what the model actually needs from this study — annual energy — and decisive for the headline capacity number, which is a reason to report both.

6.3 The eligible land is fragmented, and that is half the answer

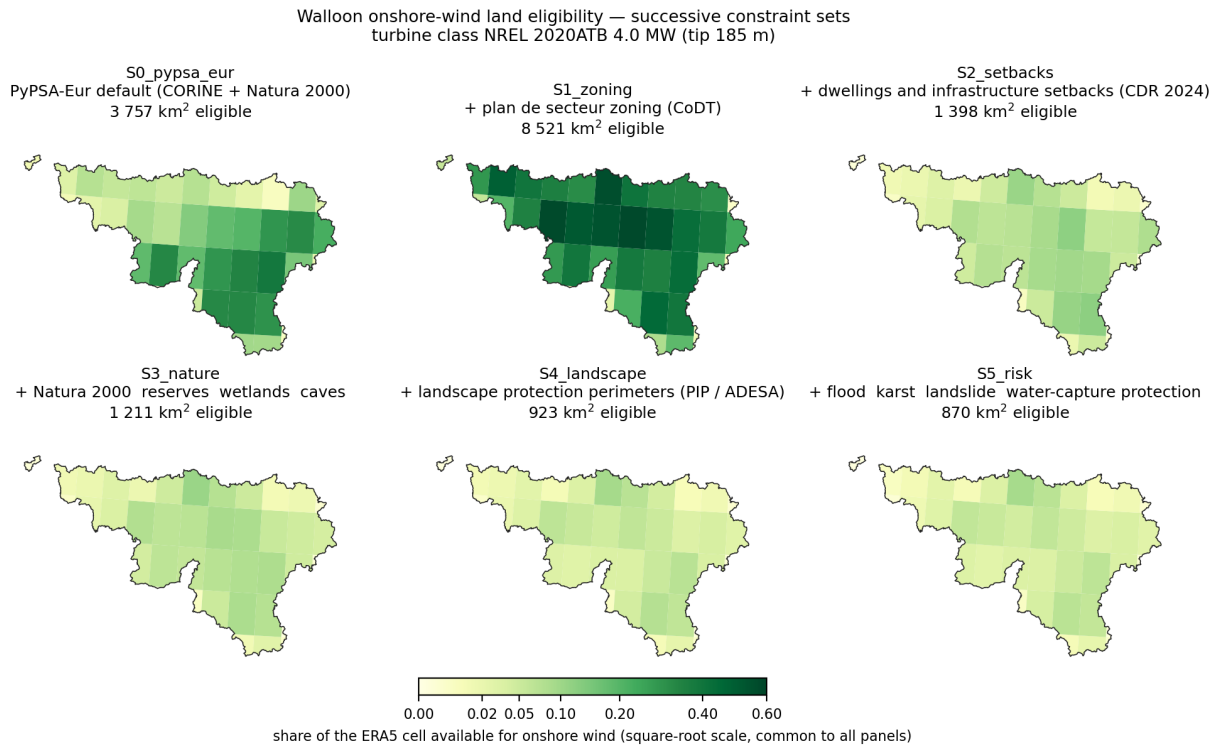
Figure 5 shows why an area figure alone is not enough. The 870 km² that survive are not a few large blocks; they are 5 290 disconnected patches with a median size of 3 ha and a largest of 8.6 km². Multiplying that area by a capacity density assumes an array at $5D \times 7D$ spacing — 0.79 km² per machine for the reference class — which a three-hectare patch cannot hold.

Screening the patches by size gives a bracket:

Neither end of the bracket is the answer. The unfiltered figure counts land no machine could occupy. The four-array filter is too strict in the other direction: a wind farm’s turbines do not have to sit inside one contiguous eligible polygon — they can straddle a road, a hedgerow or a setback sliver, and the access tracks and cabling that join them cross ineligible land as a matter of course. The 500 m-scale gaps that fragment this raster are mostly setback slivers, not obstacles.

Table 4: Eligible area after each constraint set.

region	scen.	constraint set	area [km ²]	share [%]	removed [km ²]
Wallonia	S0 pypsa eur	PyPSA-Eur default (CORINE + Natura 2000)	3 756	22.22	n/a
	S1 zoning	+ plan de secteur zoning (CoDT)	8 521	50.40	8 385
	S2 setbacks	+ dwellings and infrastructure setbacks (CDR 2024)	1 398	8.27	7 122
	S3 nature	+ Natura 2000, reserves, wetlands, caves	1 211	7.16	187
	S4 landscape	+ landscape protection perimeters (PIP / ADESA)	923	5.46	288
	S5 risk	+ flood, karst, landslide, water-capture protection	870	5.14	54
BEWAL node	S0 pypsa eur	PyPSA-Eur default (CORINE + Natura 2000)	3 597	23.74	n/a
	S1 zoning	+ plan de secteur zoning (CoDT)	7 122	47.01	8 028
	S2 setbacks	+ dwellings and infrastructure setbacks (CDR 2024)	1 309	8.64	5 812
	S3 nature	+ Natura 2000, reserves, wetlands, caves	1 123	7.41	186
	S4 landscape	+ landscape protection perimeters (PIP / ADESA)	852	5.63	271
	S5 risk	+ flood, karst, landslide, water-capture protection	813	5.36	40

**Figure 3:** Spatial effect of each constraint set. The shading is the share of each ERA5 cell that survives; the ERA5 grid is 0.3°, so these panels show *where* the resource is, not the parcel-level pattern.

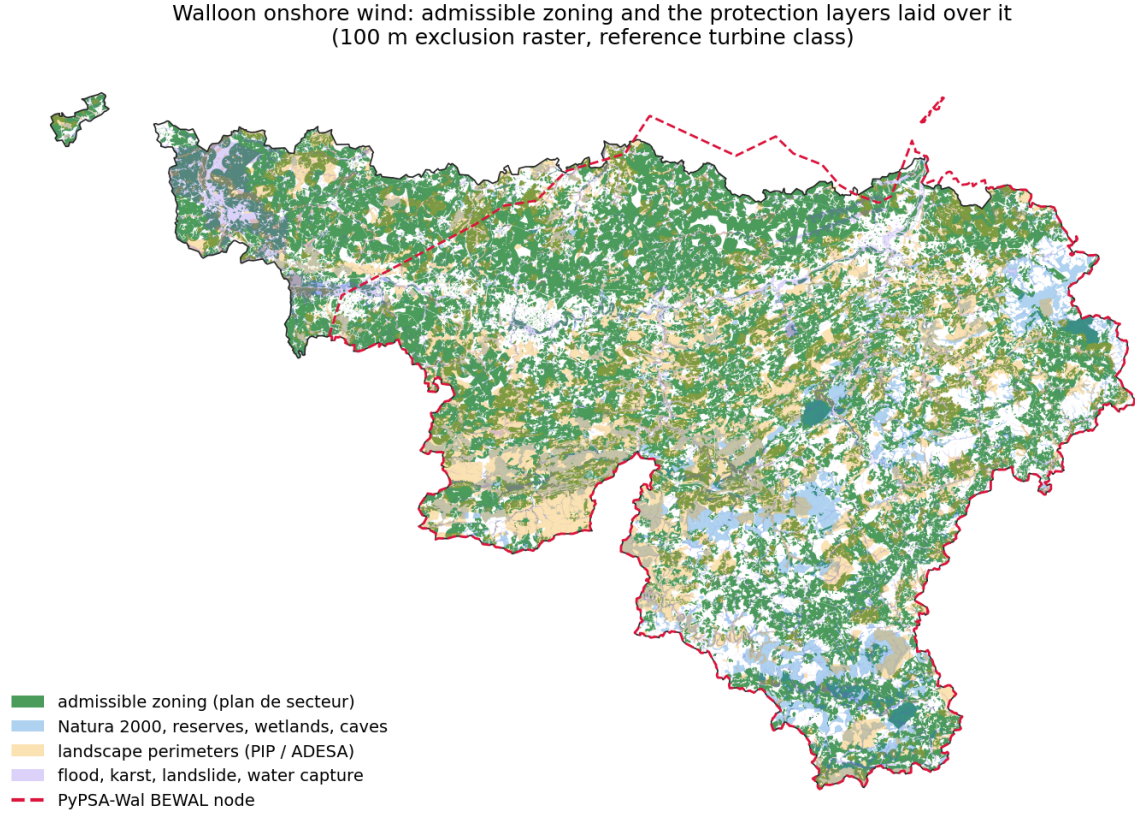


Figure 4: The admissible plan-de-secteur envelope with the protection layers laid over it, at the 100 m resolution of the exclusion raster.

Table 5: Reference scenario S5_risk across turbine classes and capacity densities. *derived* is $P_{\text{nom}}/(5D \times 7D)$; *pypsa eur* is PyPSA-Eur’s 3 MW km⁻²; *dense* is a $4D \times 6D$ upper bound.

region	turbine	density case	[MW km ⁻²]	area [km ²]	$p_{\text{nom,max}}$ [MW]	FL
Wallonia	V112 / 3.0 MW (tip 136 m)	dense	8.00	945	7 560	
	V112 / 3.0 MW (tip 136 m)	derived	6.83	945	6 457	
	V112 / 3.0 MW (tip 136 m)	pypsa eur	3.00	945	2 835	
	NREL 2020ATB 4.0 MW (tip 185 m)	dense	8.00	870	6 957	
	NREL 2020ATB 4.0 MW (tip 185 m)	derived	5.08	870	4 417	
	NREL 2020ATB 4.0 MW (tip 185 m)	pypsa eur	3.00	870	2 609	
	NREL 2020ATB 5.5 MW (tip 208 m)	dense	8.00	830	6 637	
	NREL 2020ATB 5.5 MW (tip 208 m)	derived	5.13	830	4 257	
	NREL 2020ATB 5.5 MW (tip 208 m)	pypsa eur	3.00	830	2 489	
BEWAL node	V112 / 3.0 MW (tip 136 m)	dense	8.00	882	7 059	
	V112 / 3.0 MW (tip 136 m)	derived	6.83	882	6 029	
	V112 / 3.0 MW (tip 136 m)	pypsa eur	3.00	882	2 647	
	NREL 2020ATB 4.0 MW (tip 185 m)	dense	8.00	813	6 501	
	NREL 2020ATB 4.0 MW (tip 185 m)	derived	5.08	813	4 127	
	NREL 2020ATB 4.0 MW (tip 185 m)	pypsa eur	3.00	813	2 438	
	NREL 2020ATB 5.5 MW (tip 208 m)	dense	8.00	776	6 204	
	NREL 2020ATB 5.5 MW (tip 208 m)	derived	5.13	776	3 979	
	NREL 2020ATB 5.5 MW (tip 208 m)	pypsa eur	3.00	776	2 327	

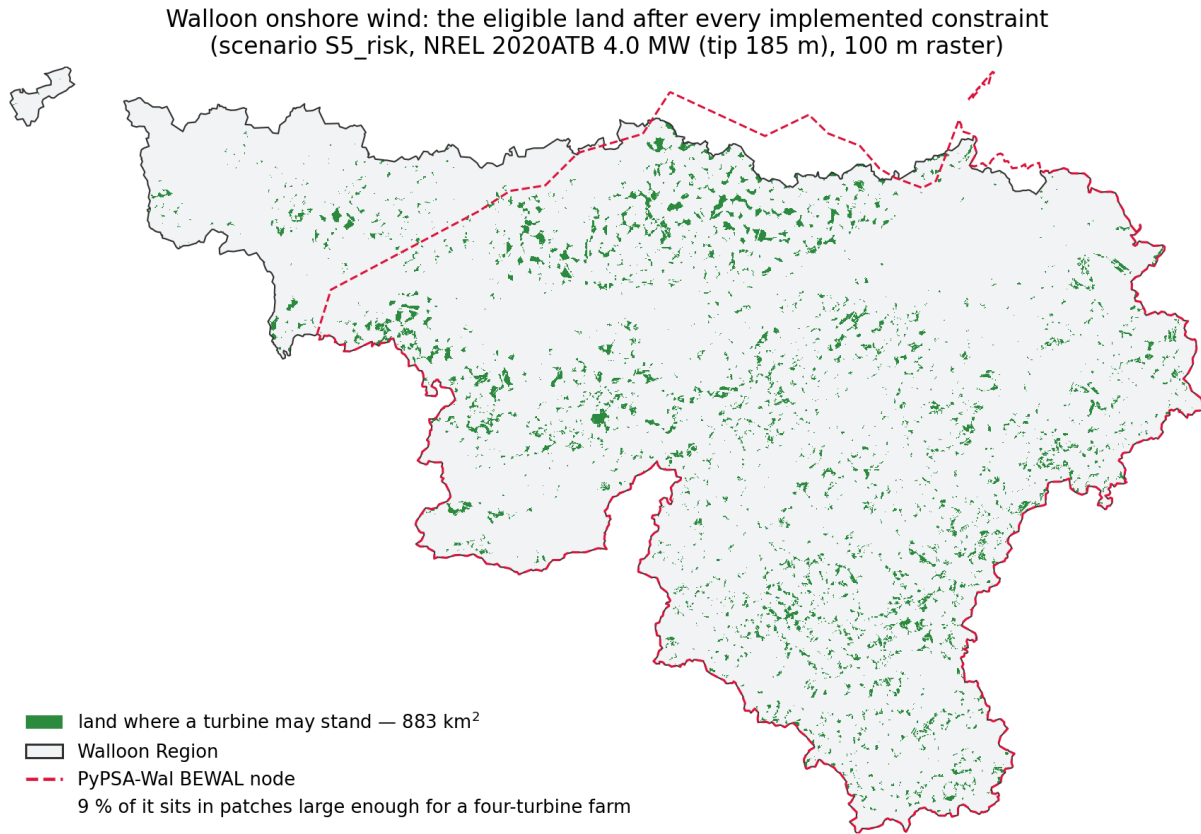


Figure 5: The answer, at 100 m resolution: the land that survives *every* implemented constraint. This is the raster the potential is integrated from, and it is written out as `results/eligible_land.tif` so it can be opened in QGIS and checked parcel by parcel.

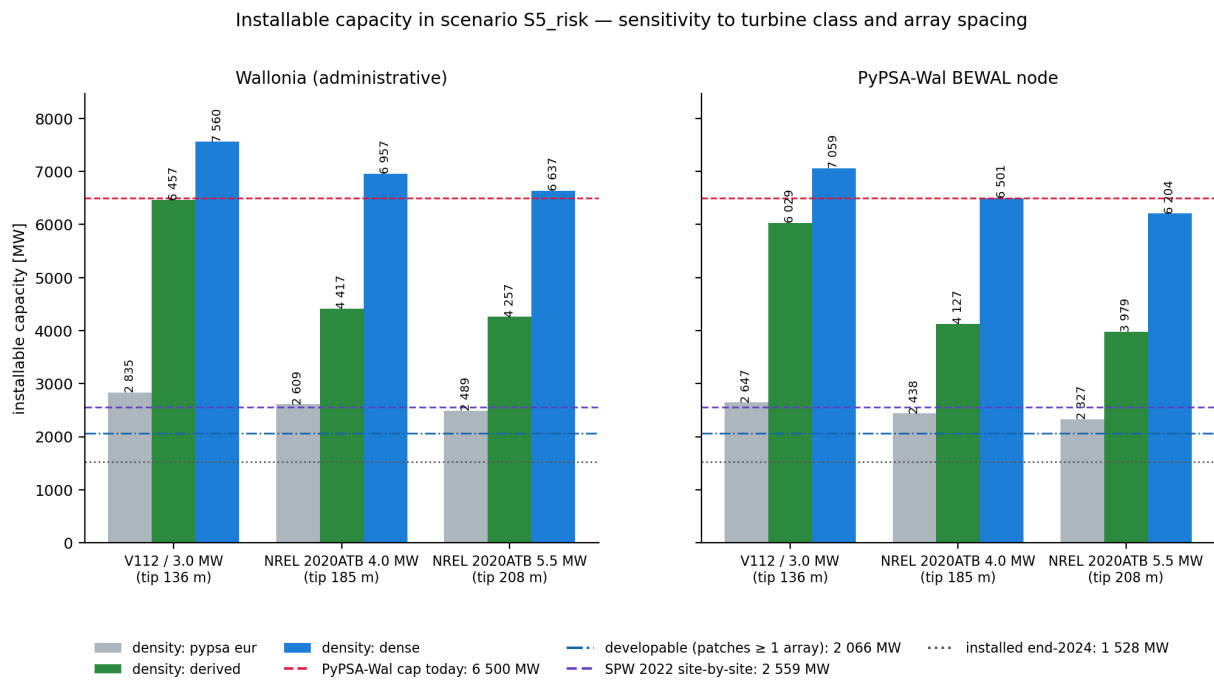


Figure 6: Installable capacity in the reference scenario, by turbine class and array spacing.

Table 6: Installable capacity under three developability screens, reference scenario and turbine class, administrative Wallonia, at 5.08 MW/km².

screen	area [km ²]	share	$p_{\text{nom,max}}$ [MW]
all eligible land	870	100 %	4 484
patches $\geq$ one turbine array	407	46 %	2 066
patches $\geq$ four turbine arrays	78	8.8 %	394

Finding. The developability screen is what reconciles this study with the Region’s own. Filtering to patches that can hold at least one turbine array gives 2 066 MW, against the 2 559 MW implied by the 457 potential turbines of the 2022 SPW/Gembloux site-by-site simulation — two methods with different data and different logic landing within 20 % of each other. That agreement is the strongest evidence in this report that the order of magnitude is right, and it puts the credible technical potential at **2–4.5 GW**, not at 6.5 GW.

6.4 Against the numbers in circulation

Table 7: This study against the figures currently used or published.

source	basis	region	P [MW]	E [TWh]
PyPSA-Wal today (input_parameters_for_models.csv)	PNEC wallon / EDORA, expert judgement	BEWAL (model)	6 500	21.14
PyPSA-Eur land eligibility (CORINE + Natura 2000)	this study, scenario S0	BEWAL (model)	18 270	58.98
Walloon legal constraint set	this study, scenario S5_risk	BEWAL (model)	4 127	13.42
Walloon legal constraint set	this study, scenario S5_risk	Wallonia (administrative)	4 417	14.37
Walloon legal constraint set, patches $\geq$ 1 turbine array	this study, S5_risk, developability filter	Wallonia (administrative)	2 066	6.72
Walloon legal constraint set, patches $\geq$ 4 turbine arrays	this study, S5_risk, strict developability filter	Wallonia (administrative)	394	1.28
SPW/Gembloux favourable-zone update 2022, '180 m' case	site-by-site simulation, additional potential only	Wallonia (administrative)	–	3.96
Installed fleet, end 2024	EDORA / Renouvelle	Wallonia (administrative)	1 528	–
PNEC / PACE 2030 onshore-wind target	Walloon Government	Wallonia (administrative)	–	6.20

Three comparisons deserve comment.

Against the 2022 SPW/Gembloux favourable-zone update. That study reports 175 km² of *zones favorables strictes* and 272 km² of *zones favorables élargies* for its “180 m” case, and 3 959 GWh/yr of *additional* production from 457 potential turbines. Two differences are structural and both are expected:

- It applies constraints this workflow cannot — aviation and defence radar, priority bird zones, classified sites, the 7 % slope criterion — so its eligible area is smaller than what is reported here for the same nominal rule set.
- It then places individual machines and screens them against landscape inter-distance and encirclement criteria, and it counts only *new* capacity, excluding the footprint of the standing fleet.

The two numbers therefore bracket the answer from opposite sides: this study is an upper bound on eligible land under the constraints it can represent; the 2022 study is a lower bound reflecting a full permitting screen on a fixed turbine layout. Both are far below 6 500 MW.

Against the PNEC/PACE target. The Walloon 2030 onshore-wind target is 6 200 GWh/yr, revised upward from the 4 600 GWh/yr of the 2019 plan — which is the figure the 2022 favourable-zone update was benchmarked against, and which it concluded was reachable. The reference case here yields 14.37 TWh/yr for the administrative Region at 3 253 full-load hours on an all-new fleet occupying every eligible parcel, which is an envelope and not a forecast.

Against the model’s own cap. 6 500 MW against the reference case’s 4 127 MW is a factor of 1.6. But the reference case is one point in a range, and 6 500 MW is inside it: the same eligible land supports 7 059 MW at the densest packing considered, and 6 029 MW on a V112-class fleet at conventional $5D \times 7D$ spacing. The honest statement is therefore narrower than “the cap is too high”.

Finding. 6 500 MW is reachable on the *unfiltered* eligible land, but only at a capacity density around 6.8 MW km^{-2} , only if the four unimplemented constraint families of §8 — radar, slope, priority bird zones, classified sites — turn out to cost little, and only if every fragment of eligible land counts — including the 54 % of it that sits in patches too small to hold a single turbine array (§6.3). Once the developability screen is applied, the two independent methods available — this one and the Region’s own 2022 site-by-site simulation — agree on 2–4.5 GW.

6.5 Full-load hours

The capacity factor is computed on the 2013 ERA5/SARAH-3 cutout with the same layout weighting PyPSA-Eur uses. For the administrative Region the reference case gives 3 253 h; for BEWAL, 3 252 h.

Those are high numbers, and the reason is the machine, not the method. The reference class is an NREL 2020 ATB 4 MW turbine with a 150 m rotor — a specific power of 226 W m^{-2} , against 305 W m^{-2} for the V112 PyPSA-Eur uses by default — on a 110 m hub instead of 80 m. Low-specific-power machines on tall towers are exactly what raises full-load hours, and it is why the same workflow returns 2 461 h when the V112 is substituted into the same eligible land (Table 5). The V112 number is the one that reconciles with §2.2 and with the model’s own resource.

Caveat. These are *new-turbine* full-load hours applied to a hypothetical all-new fleet, and they are not comparable with the 1 885 h TIMES-WAL uses for the blended existing fleet, nor with what the standing Walloon fleet actually delivers. That discrepancy is analysed in `docs/renewable-potentials.md` §10 and is not re-opened here. The relevant point for this report is that a *technical potential* is by definition about what could be built from now on, so the new-turbine figure is the right one in this context.

Caveat. Only the 2013 weather year is available on this machine. The Walloon onshore-wind resource in 2013 is about 5 % *better* than in 2010, the year the main model runs use (`docs/renewable-potentials.md` §10.1). Capacity is unaffected — it depends only on eligible area — but the energy column is optimistic by roughly that margin relative to a 2010 run.

7 Assumptions

The result depends on the following choices. Each is a config entry and each can be varied.

Capacity density. The reference is 5.08 MW/km^2 , derived from the reference turbine at $5D$ crosswind $\times 7D$ downwind spacing. This is a *technical* density on land that has already been filtered to genuinely developable parcels; PyPSA-Eur’s 3 MW km^{-2} is a deliberately conservative continental prior applied to much less filtered land. Both are reported.

Array spacing. $5D \times 7D$ is conventional for a prevailing-wind regime. A $4D \times 6D$ packing raises the density by about 45 %.

Habitat setback rule. The 2024 framework ($500 + H/2$). The 2013 rule ($4H$) is available and is more restrictive for machines above 250 m tip height — i.e. for every class considered here.

Scattered dwellings. 400 m around every ICAR address point. The address layer includes non-residential buildings (schools, churches, commercial premises), so this is mildly conservative; the 2022 SPW study makes the same approximation and notes the same caveat.

Broad-leaved / coniferous split. Taken from CORINE at 250 m; mixed stands are counted as broad-leaved. The Region’s own CARTOFOR species-proportion map would be better and is the natural upgrade (§8).

Runoff flood hazard excluded from the flood layer. The *ruissellement* hazard layer holds 5.2 million polygons and maps a thin-sheet phenomenon that does not preclude a turbine foundation. The overflow extent at the extreme return period is used instead.

Geometry generalisation. Heavy polygon layers are requested from the server with 25 m generalisation — a quarter of the exclusion raster cell, so it cannot change the aggregate.

Existing wind farms are not excluded. The result is a *gross* potential: the standing 1 528 MW occupies land that is counted as available. This is the right convention for `p_nom_max`, which is a cap on the *total* fleet, not on additions.

Every fragment of eligible land is counted in the headline figure. The developability screen of §6.3 is reported separately rather than applied, because its cut-off is a judgement.

No grid-connection or economic screen. Distance to the HV network, connection cost and wind-speed profitability thresholds are not applied. The 2022 SPW study applies a wind-resource threshold; this one does not, so its eligible area is larger on that count too.

8 Open questions

These are the gaps that stand between this analysis and one that the administration could adopt without qualification. They are listed in the order in which they would change the number.

1. **Aviation and defence radar exclusions are missing, and they are large.** In Belgium the secondary-radar exclusion radius is of the order of 10 km with a study zone out to 16 km, plus beacon and military-airspace constraints covering the whole territory. Neither skeyes nor Defence publishes the geometry as open data. Obtaining those layers — they exist, the SPW favourable-zone map uses them — is the single most important improvement, and it can only reduce the potential. *Action: request the layers from SPW Énergie or from the authors of the favourable-zone cartography.*
2. **The 7 % slope criterion is not applied.** The 2013 Walloon map excludes slopes above 7 % as technically incompatible. Only the $> 30^\circ$ layer is public here. On a 1 m LiDAR model a 7 % threshold is very restrictive; on a 30 m DEM it is a different quantity entirely. The Region’s own slope raster should be used, at a resolution the two studies agree on. *Direction: reduces the potential, most in the Ardennes.*
3. **Priority ornithological zones and classified sites are missing.** The DEMNA high-priority bird layer and the classified-site inventory are integral exclusions in the Walloon method. *Direction: reduces.*
4. **Partial constraints are not modelled at all.** The Walloon method distinguishes integral from *partial* exclusions (medium-priority bird zones, bat interest zones, migration corridors, the principal ecological structure, radar study zones at 8–16 km) and applies a 25 % success rate to turbines falling in them. Implementing them would require a site-level model rather than an area calculation. *Direction: reduces.*

5. **Landscape inter-distance and encirclement are not modelled.** These are the criteria that cut the 2022 study from 1 323 simulated turbines to 457 potential ones. An area-based method cannot represent them. *Direction: reduces, substantially.*
6. **The developability screen of §6.3 is a proxy, and it is the widest uncertainty in this report.** Requiring a patch to hold a whole $5D \times 7D$ array is too strict, because real turbines straddle setback slivers and their access tracks cross ineligible land; counting every fragment is too lax. The bracket is 2 066–4 417 MW and the right answer is inside it. Closing it properly means placing machines, i.e. doing what the 2022 study did — which is the natural next step if the cabinet wants a single number rather than a range. *Direction: the true value is below the unscreened figure.*
7. **The CORINE forest split should be replaced by CARTOFOR.** The Region’s own species-proportion map, at the resolution the plan-de-secteur forest polygons deserve, would remove a real source of error in the single constraint that matters most. *Direction: unknown.*
8. **The model’s BEWAL node should be corrected.** See Section 5. Until it is, the Walloon cap is being applied to a region that is 89.6 % of Wallonia.
9. **Repowering is not represented.** EDORA’s 3.8 GW additional potential by 2050 explicitly includes repowering of existing sites with larger machines. A land-eligibility study measures the envelope, not the turnover of the fleet inside it; the two are complementary, and the difference between them is part of why 6 500 MW and 4 127 MW differ.
10. **The weather year is 2013, not 2010.** Only capacity factors are affected; see §6.4.
11. **Should `p_nom_max` be a technical or a deployment limit?** PyPSA-Wal already carries a build-rate limit and a 2030 corridor (`docs/renewable-potentials.md` §3–4) to represent deployment. If those are doing their job, `p_nom_max` should be the technical potential and the present 6 500 MW is the wrong kind of number in the wrong slot. That is a modelling decision, not a GIS one.

9 What we recommend deciding

1. **Do not change `p_nom_max` yet.** The missing radar and slope layers all push in the same direction — downwards — so the figure reported here is an upper bound on an upper bound. Changing the cap on the basis of an incomplete constraint set would replace one unsourced number with another. What can be done now, at no risk, is to *document* the existing 6 500 MW with the land basis and capacity density that make it arithmetically reachable (870 km^2 at 6.8 MW km^{-2}), so the administration can contest a number rather than a citation.
2. **Request the missing layers** (item 1 and item 2 of §8) from SPW Énergie. They exist; this is an administrative step, not a research one.
3. **Settle the region question** (§5) before any number is quoted externally.
4. **If a single number is needed, place the machines.** The developability bracket (§6.3) is the widest remaining uncertainty and an area method cannot close it. The eligible raster is written out as `results/eligible_land.tif` precisely so that a siting pass can be run on it.
5. **Decide what `p_nom_max` is meant to be** (item 10). Until that is settled the comparison between 6 500 MW and 4 127 MW is a category error, not a discrepancy.

A Full results

B Data provenance

All layers below were retrieved from the ESRI-REST services of the Géoportail de la Wallonie (<https://geoservices.wallonie.be>) under the CC-BY 4.0 licence.

Additional inputs: ERA5/SARAH-3 `atlite` cutout for 2013 (Copernicus); CORINE Land Cover 2006 at 250 m, PyPSA-Eur data bundle; NUTS-3 shapes and the PyPSA-Wal BEWAL onshore

Table 8: All scenarios, turbine classes and regions at the reference capacity density.

region	scenario	turbine	area [km ²]	P [MW]	FLH [h]	E [TW h]
admin	S0 pypsa eur	V112 / 3.0 MW (tip 136 m)	3 757	25 668	2 442	62.67
admin	S1 zoning	V112 / 3.0 MW (tip 136 m)	8 521	58 223	2 464	143.44
admin	S2 setbacks	V112 / 3.0 MW (tip 136 m)	1 511	10 325	2 454	25.34
admin	S3 nature	V112 / 3.0 MW (tip 136 m)	1 316	8 990	2 460	22.11
admin	S4 landscape	V112 / 3.0 MW (tip 136 m)	1 004	6 859	2 459	16.87
admin	S5 risk	V112 / 3.0 MW (tip 136 m)	945	6 457	2 461	15.89
admin	S0 pypsa eur	NREL 2020ATB 4.0 MW (tip 185 m)	3 757	19 079	3 230	61.62
admin	S1 zoning	NREL 2020ATB 4.0 MW (tip 185 m)	8 521	43 277	3 257	140.96
admin	S2 setbacks	NREL 2020ATB 4.0 MW (tip 185 m)	1 398	7 103	3 244	23.04
admin	S3 nature	NREL 2020ATB 4.0 MW (tip 185 m)	1 211	6 152	3 251	20.00
admin	S4 landscape	NREL 2020ATB 4.0 MW (tip 185 m)	923	4 689	3 251	15.25
admin	S5 risk	NREL 2020ATB 4.0 MW (tip 185 m)	870	4 417	3 253	14.37
admin	S0 pypsa eur	NREL 2020ATB 5.5 MW (tip 208 m)	3 757	19 275	3 302	63.65
admin	S1 zoning	NREL 2020ATB 5.5 MW (tip 208 m)	8 521	43 720	3 331	145.63
admin	S2 setbacks	NREL 2020ATB 5.5 MW (tip 208 m)	1 341	6 879	3 317	22.82
admin	S3 nature	NREL 2020ATB 5.5 MW (tip 208 m)	1 157	5 937	3 324	19.73
admin	S4 landscape	NREL 2020ATB 5.5 MW (tip 208 m)	881	4 518	3 325	15.02
admin	S5 risk	NREL 2020ATB 5.5 MW (tip 208 m)	830	4 257	3 326	14.16
model	S0 pypsa eur	V112 / 3.0 MW (tip 136 m)	3 597	24 579	2 442	60.02
model	S1 zoning	V112 / 3.0 MW (tip 136 m)	7 122	48 663	2 467	120.07
model	S2 setbacks	V112 / 3.0 MW (tip 136 m)	1 413	9 658	2 455	23.70
model	S3 nature	V112 / 3.0 MW (tip 136 m)	1 219	8 329	2 461	20.50
model	S4 landscape	V112 / 3.0 MW (tip 136 m)	926	6 325	2 461	15.56
model	S5 risk	V112 / 3.0 MW (tip 136 m)	882	6 029	2 462	14.84
model	S0 pypsa eur	NREL 2020ATB 4.0 MW (tip 185 m)	3 597	18 270	3 228	58.98
model	S1 zoning	NREL 2020ATB 4.0 MW (tip 185 m)	7 122	36 172	3 256	117.77
model	S2 setbacks	NREL 2020ATB 4.0 MW (tip 185 m)	1 309	6 650	3 242	21.56
model	S3 nature	NREL 2020ATB 4.0 MW (tip 185 m)	1 123	5 703	3 250	18.53
model	S4 landscape	NREL 2020ATB 4.0 MW (tip 185 m)	852	4 328	3 251	14.07
model	S5 risk	NREL 2020ATB 4.0 MW (tip 185 m)	813	4 127	3 252	13.42
model	S0 pypsa eur	NREL 2020ATB 5.5 MW (tip 208 m)	3 597	18 457	3 301	60.92
model	S1 zoning	NREL 2020ATB 5.5 MW (tip 208 m)	7 122	36 542	3 329	121.64
model	S2 setbacks	NREL 2020ATB 5.5 MW (tip 208 m)	1 256	6 444	3 315	21.36
model	S3 nature	NREL 2020ATB 5.5 MW (tip 208 m)	1 073	5 506	3 322	18.29
model	S4 landscape	NREL 2020ATB 5.5 MW (tip 208 m)	813	4 172	3 324	13.87
model	S5 risk	NREL 2020ATB 5.5 MW (tip 208 m)	776	3 979	3 325	13.23

Table 9: Source datasets, as retrieved.

dataset	service / layer	features	retrieved
Points d’adresse ICAR	DONNEES_BASE/ICAR_- ADR_PT/1	1 512 246	2026-09-22
Périmètres d’intérêt paysager de l’inventaire ADESA	AMENAGEMENT_TERRI- TOIRE/ADESA/1	1 142	2026-09-22
Cavités souterraines d’intérêt scien- tifique	FAUNE_- FLORE/CONSNAT/0	80	2026-09-22
Zones inondables — débordement, péri- ode de retour extrême	EAU/ZI/2	39 893	2026-09-22
Zones de contraintes karstiques	AMENAGEMENT_TERRI- TOIRE/CONTR_KARST/0	449	2026-09-22
Contraintes physiques liées aux éboule- ments	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/0	997	2026-09-22
Réseau Natura 2000 — périmètres des sites en vigueur	FAUNE_- FLORE/NATURA2000/9	240	2026-09-22
Plan de secteur — lignes électriques haute tension	AMENAGEMENT_TERRI- TOIRE/PDS/12	1 016	2026-09-22
Plan de secteur — périmètres d’intérêt paysager	AMENAGEMENT_TERRI- TOIRE/PDS/17	1 456	2026-09-22
Plan de secteur — réseau ferroviaire	AMENAGEMENT_TERRI- TOIRE/PDS/11	911	2026-09-22
Plan de secteur — réseau routier	AMENAGEMENT_TERRI- TOIRE/PDS/9	5 420	2026-09-22
Plan de secteur — voies navigables	AMENAGEMENT_TERRI- TOIRE/PDS/13	1	2026-09-22
Plan de secteur — zones d’affectation	AMENAGEMENT_TERRI- TOIRE/PDS/22	43 797	2026-09-22
Réserves naturelles agréées	FAUNE_- FLORE/CONSNAT/3	264	2026-09-22
Réserves naturelles domaniales	FAUNE_- FLORE/CONSNAT/2	306	2026-09-22
Réserves forestières	FAUNE_- FLORE/CONSNAT/1	20	2026-09-22
Versants supérieurs à 30 degrés	AMENAGEMENT_TERRI- TOIRE/RISQ_EBOULT/1	2 834	2026-09-22
Zones de prévention rapprochée (IIa) des captages	EAU/PROTECT_CAPT/2	736	2026-09-22
Zones humides d’intérêt biologique	FAUNE_- FLORE/CONSNAT/4	73	2026-09-22

region, PyPSA-Eur resources; turbine power curves from the `atlite` library (Vestas, NREL 2020 ATB).